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EXPERIMENT 3(D)
COMPARISON OF UCI AND PIMA DIABETES DATASETS
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UCI DIABETES DATASET

	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	182	73	37	215	41.81
1	131	90	34	153	34.91
2	172	74	48	133	25.34
3	94	67	42	276	25.90
4	186	73	10	161	36.85

	DiabetesPedigreeFunction	Age	Outcome
0	1.879	69	1
1	1.422	48	0
2	1.183	52	0
3	0.285	47	1
4	1.269	40	1

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Shape:
(100, 8)
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PIMA INDIANS DIABETES DATASET

	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	88	70	28	258	30.02
1	104	83	48	265	36.75
2	147	63	27	262	30.64
3	183	101	20	151	23.29
4	167	103	23	110	34.06

	DiabetesPedigreeFunction	Age	Outcome
0	0.884	43	1
1	0.463	39	0
2	1.441	55	1
3	0.557	73	0
4	1.718	40	1

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Shape:
(100, 8)
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UNIVARIATE ANALYSIS - UCI

	Mean	Median	Variance	Standard Deviation \
Glucose	137.2900	139.0000	1161.7433	34.0844
BloodPressure	83.5000	84.5000	180.2727	13.4266
SkinThickness	29.3600	30.0000	155.0206	12.4507
Insulin	175.3300	177.0000	5195.0920	72.0770
BMI	31.1328	31.1750	21.5920	4.6467
DiabetesPedigreeFunction	1.0335	1.0885	0.2851	0.5339
Age	50.1300	50.5000	230.7405	15.1901

	Minimum	Maximum
Glucose	81.000	196.000
BloodPressure	60.000	104.000
SkinThickness	10.000	49.000
Insulin	53.000	297.000
BMI	18.010	41.810
DiabetesPedigreeFunction	0.127	1.981
Age	21.000	79.000

UNIVARIATE ANALYSIS - PIMA

	Mean	Median	Variance	Standard Deviation \
Glucose	136.9500	136.5000	1210.0076	34.7852
BloodPressure	82.1600	83.0000	161.3277	12.7015
SkinThickness	30.7300	29.0000	143.4516	11.9771
Insulin	176.0500	175.0000	4657.8460	68.2484
BMI	32.5641	32.1150	24.9800	4.9980
DiabetesPedigreeFunction	1.0089	1.0125	0.3171	0.5631

UNIVARIATE ANALYSIS - PIMA

	Mean	Median	Variance	Standard Deviation
Glucose	136.9500	136.5000	1210.0076	34.7852
BloodPressure	82.1600	83.0000	161.3277	12.7015
SkinThickness	30.7300	29.0000	143.4516	11.9771
Insulin	176.0500	175.0000	4657.8460	68.2484
BMI	32.5641	32.1150	24.9800	4.9980
DiabetesPedigreeFunction	1.0089	1.0125	0.3171	0.5631
Age	49.8200	48.0000	265.5834	16.2967

	Minimum	Maximum
Glucose	80.000	197.000
BloodPressure	60.000	104.000
SkinThickness	10.000	49.000
Insulin	50.000	296.000
BMI	19.620	46.910
DiabetesPedigreeFunction	0.119	1.985
Age	21.000	79.000

MEAN COMPARISON

	UCI Mean	Pima Mean
Glucose	137.2900	136.9500
BloodPressure	83.5000	82.1600
SkinThickness	29.3600	30.7300
Insulin	175.3300	176.0500
BMI	31.1328	32.5641
DiabetesPedigreeFunction	1.0335	1.0089
Age	50.1300	49.8200

FINAL COMPARISON OF ANALYSIS RESULTS

	Analysis	UCI Diabetes	Pima Indians
Linear Regression R ²		0.1753	0.0362
Logistic Regression Accuracy		0.3500	0.5500
Multiple Regression R ²		0.2133	0.0502

DATASET CHARACTERISTICS

UCI average Glucose : 137.29
Pima average Glucose: 136.95
UCI average BMI : 31.13
Pima average BMI : 32.56

INTERPRETATION

1. Univariate analysis compares the central tendency and dispersion of variables.
2. Bivariate analysis compares the relationship between variables and classification performance.
3. Multiple regression compares the ability to predict BMI using multiple independent variables.
4. UCI shows a higher Multiple Regression R² than the Pima dataset.
5. Differences in the statistical and regression results reflect differences in the distributions and relationships within the two datasets.

RESULT

The statistical analysis results of the UCI Diabetes and Pima Indians Diabetes datasets were successfully compared. Univariate statistics, bivariate regression results, and multiple regression performance were analyzed.

EXPERIMENT 3(D) COMPLETED SUCCESSFULLY